Seat No: ___________ **Enrollment No: ______________**

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech. Summer 2017 - 18 Examination

Semester: 3 **Date: 11/06/2018**
Subject Code: 03105202 **Time: 02:00pm to 04:30 pm**
Subject Name: Operating System **Total Marks: 60**

Instructions:
1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 Objective Type Questions - (All are compulsory) (Each of one mark) **(15)**

1. Operating system is a____________.
   a) System software
   b) Application software
   c) Presentation software
   d) Database software.
2. In producer-consumer problem, when buffer status is empty _____ has to wait.
   a) Producer
   b) Consumer
   c) None
   d) Both.
3. If the resources are always preempted from the same process, we use _____ technique.
   a) Deadlock
   b) Aging
   c) System crash
   d) Starvation.
4. What is compaction?
   a) technique for overcoming internal fragmentation
   b) paging technique
   c) technique for overcoming external fragmentation
   d) technique for overcoming fetal error.
5. Operating System maintains the PCB for:
   a) each process
   b) each thread
   c) each instruction
   d) each address.
6. Differentiate between Assembler and Interpreter .
7. List out different page replacement policies in memory management.
8. What is a Time Sharing System?
9. Difference between Fixed storage and variable storage.
10. Define semaphore and what are its uses?
11. A system call to create a child process is ________
12. A process running in background is called _______________.
13. Suppose that process is in BLOCK state and waiting for some I/O services when service is completed, it goes to the _____________ state.
14. ______ is called lightweight process.
15. High page fault leads to _____

Q.2 Answer the following questions. (Attempt any three) **(15)**

A) Write different Operating System Services with diagram.

B) Define process. Differentiate between a process and a program. Explain different states of a process with a suitable diagram.

C) List out different types of Semaphore. Explain Producer-Consumer Problem.

D) What is Deadlock? List the conditions that lead to deadlock. Explain any two ways to get recovery from Deadlock.

Q.3 A) Explain the following in detail with suitable diagrams: **(07)**

(i) Swapping
(ii) Segmentation.
(iii) Multiprogramming with fixed partitions
(iv) Multiprogramming with Variable Partitions

B) What is TLB? How does it help to speed up paging? **(08)**

OR

B) Explain the various file allocation methods. **(08)**

Q.4 A) i) Explain the Virus and Malware systems. **(07)**

ii) Explain the goals of I/O software.

OR

A) Suppose Disk drive has 200 cylinders. The current position of head is 53. The queue of pending request is 98, 183, 37, 122, 14, 124, 65, 67.The Head pointer is at 53. Calculate head movement for the following algorithms. **(07)**

(i) FCFS
(ii) SSTF

B) For the following page reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 Calculate the page faults applying the following Page Replacement Algorithms for a memory with three frames: **(08)**

(i) FIFO
(ii) Optimal
(iii) LRU

Explain the above three techniques.